

LUTHANDO MBUYANE

Cape Town, South Africa | Available Immediately | Open to Relocation

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GitHub: <https://github.com/nonluthando> | Portfolio: <https://nonluthando.github.io/portfolio-v2>

PROFESSIONAL SUMMARY

Computer Science graduate from the University of Cape Town with hands-on experience building backend systems, data-driven applications, and production integrations using Python, FastAPI, PostgreSQL, Laravel, SQL, and analytics tooling. Experienced contributing to live production environments, API integrations, automated testing, data workflows, and modular application development. Strong analytical foundation through Computer Science, Statistics, and Psychology, with growing specialization toward data engineering and AI-powered systems.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java, PHP
- **Backend & APIs:** FastAPI, Flask, Laravel, REST APIs
- **Databases:** PostgreSQL, MySQL, MongoDB, DuckDB
- **Data & Analytics:** Pandas, Data Cleaning, Data Validation, Statistical Analysis, ETL Concepts, Data Visualisation
- **Testing & Dev Tools:** pytest, PHPUnit, Git, GitHub Actions, Docker
- **Cloud / AI:** AWS, Render, spaCy, Named Entity Recognition (NER)

EXPERIENCE

Software Engineering Contractor | Fixed-Term Contract

March 2026 – May 2026

- Developed backend functionality using PHP/Laravel across multiple modules within a production application.
- Built and maintained modular packages integrated into the main Laravel codebase, supporting release workflows and dependency updates.
- Integrated external inference APIs and contributed to spaCy Named Entity Recognition (NER) workflows through training, testing, and evaluation.
- Wrote unit and integration tests using PHPUnit and supported debugging, dependency troubleshooting, and production investigations.

EDUCATION

University of Cape Town (UCT) | Bachelor of Science (BSc) — Computer Science & Psychology | 2026

Relevant Coursework: Data Structures & Algorithms, Databases, Computer Networks, Operating Systems, Statistical Inference, Experimental Design, Operations Research

Awards: Dean's Merit List

PROJECTS

ScoreRent — Rental Decision Support Platform

Live Demo: <https://scorerent.onrender.com>

Python • FastAPI • PostgreSQL • Docker • GitHub Actions

- Designed and deployed a decision-support platform enabling renters to evaluate property affordability and application risk before paying non-refundable application fees.
- Developed authenticated renter profiles, evaluation history tracking, PostgreSQL persistence, and modular backend architecture.
- Implemented automated testing with pytest and CI pipelines using GitHub Actions while managing deployment debugging and cloud configuration workflows.

CSV Data Cleaning & Conversion Pipeline

Python • Pandas • OpenPyXL • CLI Tooling

- Built a modular data-preparation pipeline to transform raw CSV datasets into clean, analysis-ready Excel outputs.
- Implemented configurable cleaning workflows for handling missing values, inconsistent formatting, schema validation, and invalid records.
- Added validation and logging mechanisms to improve reproducibility, debugging, and downstream data reliability.

Data Analytics & SQL Projects

Python • SQL • DuckDB • Pandas

- Built analytical workflows using SQL, DuckDB, and Python to explore retail, sales, and behavioural datasets across multiple case studies.
- Performed customer segmentation, funnel analysis, regional sales aggregation, and trial-vs-control evaluation using structured query pipelines and statistical reasoning.
- Developed business-focused insights and recommendations by analysing transaction trends, conversion bottlenecks, and customer behaviour patterns.

SeaClear — Beach Water Quality Platform

Python • Flask • MongoDB • Agile • HCI

- Contributed to the development of a beach water quality platform focused on accessibility, usability, and public information delivery.
- Improved navigation, content workflows, and user experience through heuristic evaluation, usability testing, and backend feature implementation.

LEADERSHIP & SERVICE

Computer Science & Psychology Tutor, Avenue Road Residence (2022)

- Supported students with programming, debugging, and analytical problem solving across Python and Java coursework.

Events Subcommittee Member, UCT Science Students Council (2025)

- Contributed to communication, coordination, and student engagement initiatives across science student communities.